



**MUMBAI EDUCATIONAL TRUST
MET INSTITUTE OF COMPUTER SCIENCE**

Practical No.	LAB 07
Roll No:	1551
Title of Program:	Hive
Objective:	Hive Commands

Commands:

- a) hive
 - b) create database EmployeeDB ;
 - c) use EmployeeDB ;
 - d) set [hive.cli.print.current.db](#)=true;
 - e) create table emp(empid int, name string, salary float) row format delimited fields terminated by ',' ;
 - f) describe emp;
 - g) load data local inpath '/home/cloudera/Desktop/emp.csv' into table emp;
 - h) select * from emp;
 - i) select * from emp order by salary;
 - j) select * from emp order by salary desc;
- a) hive

```
The Edit View Search Terminal Help
[cloudera@quickstart ~]$ hive

Logging initialized using configuration in jar:file:/usr/lib/hive/lib/hive-common-1.1.0-cdh5.13.0.jar!/hive-log4j.properties
WARNING: Hive CLI is deprecated and migration to Beeline is recommended.
```

- b) create database EmployeeDB ;

```
hive> show databases;
OK
default
student
Time taken: 1.831 seconds, Fetched: 2 row(s)
hive> create database EmployeeDB ;
OK
Time taken: 0.605 seconds
```

- c) Use EmployeeDB;

```
hive> use EmployeeDB;
OK
Time taken: 0.031 seconds
hive> █
```

- d) set [hive.cli.print.current.db=true](#);
e) create table emp(empid int, name string, salary float) row format delimited fields terminated by ',';

```
hive> set hive.cli.print.current.db=true ;
hive (EmployeeDB)> create table emp (empid int, name string, salary float) row f
ormat delimited fields terminated by ',' ;
OK
Time taken: 0.258 seconds
hive (EmployeeDB)> █
```

- f) describe emp;

```
hive (EmployeeDB)> describe emp;
OK
empid                int
name                 string
salary              float
Time taken: 0.146 seconds, Fetched: 3 row(s)
hive (EmployeeDB)> █
```

- g) load data local inpath '/home/cloudera/Desktop/emp.csv' into table emp;

```
hive> load data local inpath '/home/cloudera/Desktop/emp.csv' into table emp;
Loading data to table employeeedb.emp
Table employeeedb.emp stats: [numFiles=1, totalSize=135]
OK
Time taken: 0.691 seconds
```

h) select * from emp;

```
hive> select * from emp;
OK
101      kaysan 10000.0
102      ajay   3500.0
103      vijay  5000.0
104      raj    5500.0
105      amit   6000.0
106      aman   7000.0
107      akshay 8000.0
108      rakesh 8600.0
Time taken: 0.28 seconds, Fetched: 8 row(s)
```

Write a query to find max salary, min salary, avg salary

select max(salary) as MaxSal, min(salary) as MinSal, avg(salary) as AvgSal from emp;

```
hive> select max(salary) as MaxSal, min(salary) as MinSal, avg(salary) as AvgSal
from emp;
Query ID = cloudera_20250811013535_4b267c9d-d0fd-409c-b2ee-39e991276025
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1754891977502_0003, Tracking URL = http://quickstart.cloudera
:8088/proxy/application_1754891977502_0003/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1754891977502_0003
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2025-08-11 01:35:20,085 Stage-1 map = 0%, reduce = 0%
2025-08-11 01:35:25,428 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 0.72 se
c
2025-08-11 01:35:33,746 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 1.65
sec
MapReduce Total cumulative CPU time: 1 seconds 650 msec
Ended Job = job_1754891977502_0003
MapReduce Jobs Launched:
```

```

MapReduce Total cumulative CPU time: 1 seconds 650 msec
Ended Job = job_1754891977502_0003
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 1.65 sec HDFS Read: 9438 HDFS Write: 22 SUCCESS
Total MapReduce CPU Time Spent: 1 seconds 650 msec
OK
10000.0 3500.0 6700.0
Time taken: 23.655 seconds, Fetched: 1 row(s)

```

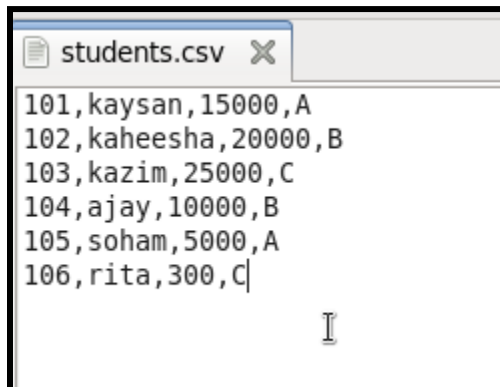
Create students table

```

hive> create table students (sid int, name string, sal float, division string) row
format delimited fields terminated by ',';
OK
Time taken: 0.142 seconds
hive>

```

Create a students.csv



```

students.csv
101,kaysan,15000,A
102,kaheesha,20000,B
103,kazim,25000,C
104,ajay,10000,B
105,soham,5000,A
106,rita,300,C

```

Load the csv data into table

```

hive> load data local inpath '/home/cloudera/Desktop/students.csv' into table s
udents;
Loading data to table student.students
Table student.students stats: [numFiles=1, totalSize=108]
OK

```

select * from students;

```
hive> select * from students;
OK
101      kaysan   15000.0 A
102      kaheesha      20000.0 B
103      kazim    25000.0 C
104      ajay     10000.0 B
105      soham     5000.0 A
106      rita      300.0   C
Time taken: 0.043 seconds, Fetched: 6 row(s)
```

Write a query to get the count : -

count(*) - Includes NULL values

count() - Does not include NULL values

select count(*) from students group by division;

```
MapReduce Total cumulative CPU time: 2 seconds 40 msec
Ended Job = job_1754891977502_0004
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 2.04 sec HDFS Read: 8127 HD
FS Write: 8 SUCCESS
Total MapReduce CPU Time Spent: 2 seconds 40 msec
OK
2
2
1
1
Time taken: 26.594 seconds, Fetched: 4 row(s)
```

Write a query for the standard deviation : -

select stddev_samp(sal) from students;

```
Total MapReduce CPU Time Spent: 1 seconds 900 msec
OK
9274.427206032726
Time taken: 24.245 seconds, Fetched: 1 row(s)
hive> █
```

Partitioning in Hive

- a) create table sales_data (pid int, cid int, amount double) partitioned by (year int) row format delimited fields terminated by ',';

```
hive> create table sales_data (pid int, cid int, amount double)partitioned by (year int) row format delimited fields terminated by ',';
FAILED: ParseException line 1:57 missing EOF at 'partitioned' near ')'
hive> create table sales_data (pid int, cid int, amount double)partitioned by (year int) row format delimited fields terminated by ',';
OK
Time taken: 0.06 seconds
```

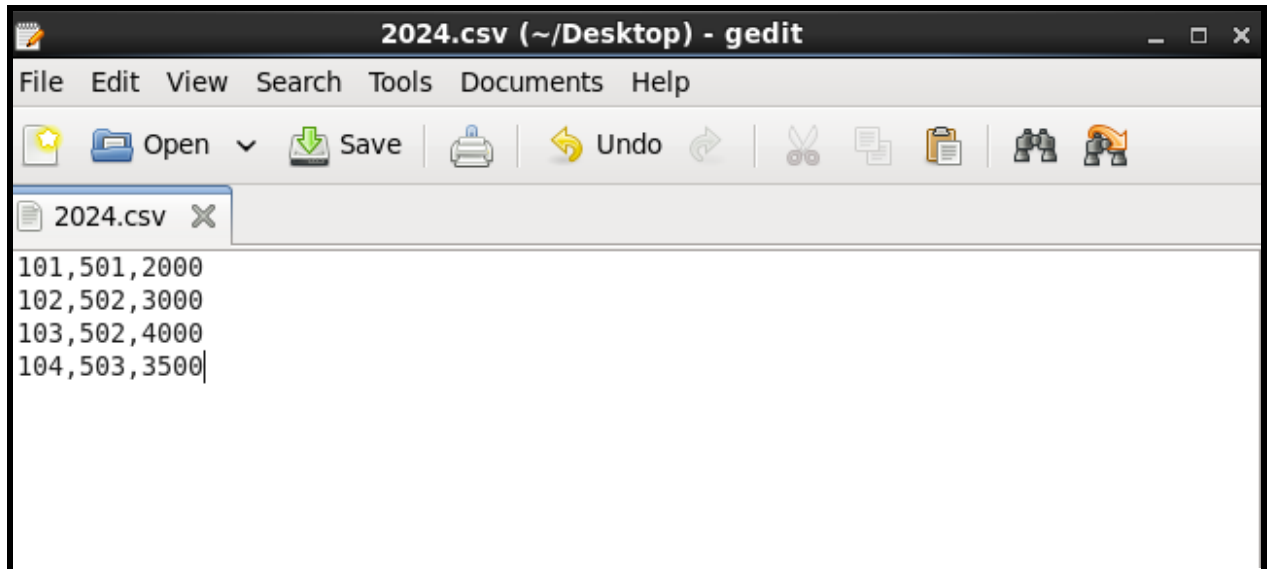
- b) describe sales_data;

```
hive> describe sales_data;
OK
pid                int
cid                int
amount             double
year               int

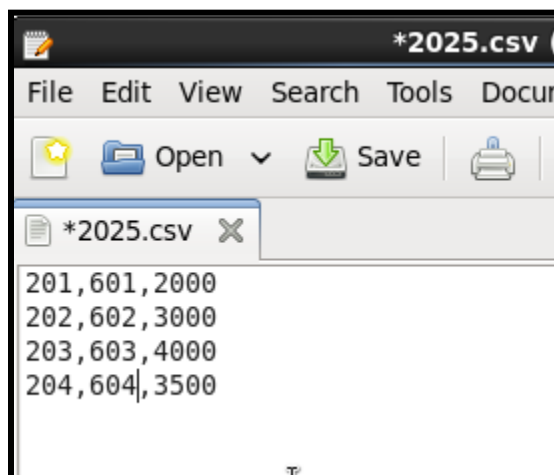
# Partition Information
# col_name          data_type          comment
year               int
Time taken: 0.153 seconds, Fetched: 9 row(s)
hive>
```

Static Partitioning

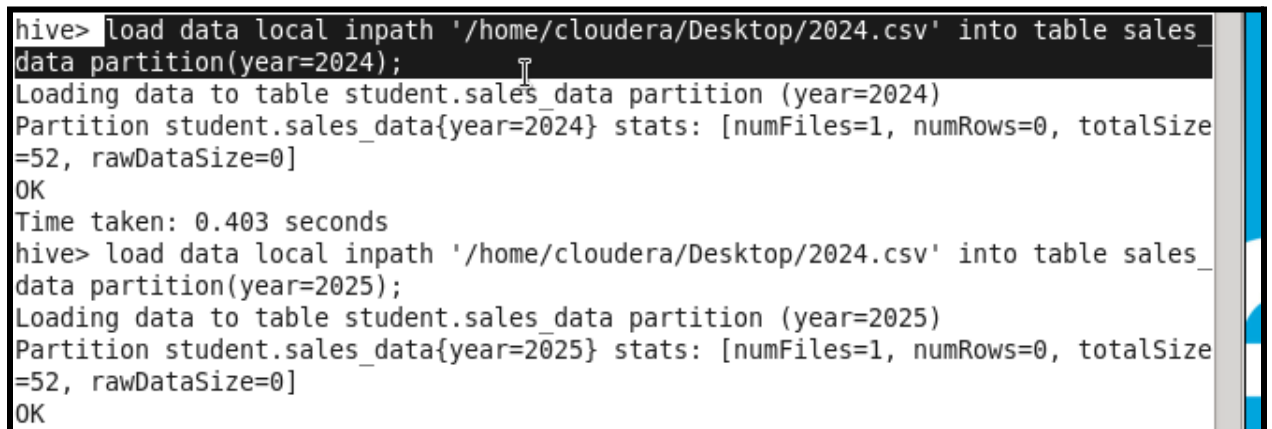
Create a 2024.csv



Create a 2025.csv



Load the data: -



select * from sales_data;

```
101      501      2000.0  2025
102      502      3000.0  2025
103      502      4000.0  2025
104      503      3500.0  2025
201      601      2000.0  2025
202      602      3000.0  2025
203      603      4000.0  2025
204      604      3500.0  2025
Time taken: 0.066 seconds, Fetched: 12 row(s)
```

desc sales_data;

```
hive> describe sales_data;
OK
pid                int
cid                int
amount             double
year               int

# Partition Information
# col_name          data_type          comment
year                int
Time taken: 0.07 seconds, Fetched: 9 row(s)
```


Dynamic Partitioning

create table student_part(sid int, name string, sal double) partitioned by (division string) row format delimited fields terminated by ',';

```
hive> create table student_part(sid int, name string, sal double) partitioned by  
(division string) row format delimited fields terminated by ',';  
OK  
Time taken: 0.09 seconds
```

Now, enter these two partitioning commands

- a) set hive.exec.dynamic.partition.mode=nostrict;
- b) set hive.exec.dynamic.partition=true;

```
hive> set hive.exec.dynamic.partition.mode=nostrict;  
hive> set hive.exec.dynamic.partition=true;
```

Insert the partition into the table:

```
hive> insert into table student_part partition(division) select sid, name, sal,  
division from students ;  
Query ID = cloudera_20250811023636_4a52c245-b952-4fc9-81ac-9f215c57c22b  
Total jobs = 3  
Launching Job 1 out of 3  
Number of reduce tasks is set to 0 since there's no reduce operator  
Starting Job = job_1754891977502_0006, Tracking URL = http://quickstart.cloudera  
:8088/proxy/application_1754891977502_0006/  
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1754891977502_0006  
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 0  
2025-08-11 02:36:53,902 Stage-1 map = 0%, reduce = 0%  
2025-08-11 02:37:02,394 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 1.27 se  
c  
MapReduce Total cumulative CPU time: 1 seconds 270 msec  
-----
```

show partitions student_part;

```
hive> show partitions student_part;  
OK  
division=A  
division=B  
division=C  
Time taken: 0.094 seconds, Fetched: 3 row(s)
```